INCH-POUND
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SUPERSEDING
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DETAIL SPECIFICATION

VALVE, SAFETY – BOILER, REMOTE PILOT OPERATED, FOR SHIPBOARD USE

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers high-performance boiler safety (unloader) valves with remote, hydraulic, non-steam flow pilot operated assembly for steam drum and superheater service at working pressures of 600, 900, and 1,500 pounds per square inch (lb/in²) for shipboard use.

1.2 Classification. Valves are of the following classes and styles, as specified (see 6.2).

1.2.1 Classes. The classes of valves are as follows (refer to 6.7 for a cross-reference of old to new class designations and substitutability data):

a. Class 1 – 600 lb/in² service
b. Class 2 – 900 lb/in² service
c. Class 3 – 1,500 lb/in² service

1.2.2 Styles. The styles of valves are as follows (see 3.3.1.2):

a. 9
b. 23
c. 25
d. 43
e. 51
f. 57

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 4820

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-H28	-	Screw-Thread Standards for Federal Services
FED-STD-H28/1	-	Screw-Thread Standards for Federal Services Section 1 Nomenclature, Definitions and Letter Symbols for Screw Threads
FED-STD-H28/2	-	Screw-Thread Standards for Federal Services Section 2 Unified Inch Screw Threads – UN and UNR Thread Forms
FED-STD-H28/4	-	Screw-Thread Standards for Federal Services Section 4 Controlled Radius Root Screw Threads, UNJ Symbol
FED-STD-H28/6	-	Screw-Thread Standards for Federal Services Section 6 Gages and Gaging for Unified Screw Threads – UN and UNR Thread Forms
FED-STD-H28/19	-	Screw-Thread Standards for Federal Services Section 19 Miscellaneous Threads
FED-STD-H28/21	-	Screw-Thread Standards for Federal Services Section 21 Metric Screw-Threads

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901	-	Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for
MIL-DTL-1222	-	Studs, Bolts, Screws and Nuts for Applications where a High Degree of Reliability is Required; General Specification for
MIL-DTL-15024	-	Plates and Tags for Identification of Equipment, General Specification for
MIL-P-15024/5	-	Plates, Identification
MIL-R-17131	-	Rods and Powders, Welding, Surfacing

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-167-1	-	Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally Excited)
MIL-STD-777	-	Schedule of Piping, Valves, Fittings, and Associated Piping Components for Naval Surface Ships
MIL-STD-1472	-	Human Engineering

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-470	-	Designing and Developing Maintainable Products and Systems, Volume I

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

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NAVAL SEA SYSTEMS COMMAND (NAVSEA) DRAWINGS

NAVSHIPS 5000-S5103-841622	-	Valves, Boiler, Safety, Dimensional Standard for 300, 600, and 1500 PSI Nominal Pressure

(Copies of this document are available from the applicable repositories listed in S0005-AE-PRO-010/EDM, which can be obtained online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/. Copies of this document may also be obtained from the Naval Ships Engineering Drawing Repository (NSEDR) online at https://199.208.213.105/webjedmics/index.jsp. To request an NSEDR account for drawing access, send an email to NNSY_JEDMICS_NSEDR_HELP_DESK@navy.mil.)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9074-AQ-GIB-010/248	-	Requirements for Welding and Brazing Procedure and Performance Qualification
S9074-AR-GIB-010/278	-	Requirements for Fabrication Welding and Inspection, and Casting Inspection and Repair for Machinery, Piping, and Pressure Vessels

(Copies of these documents are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/ by searching for the TMIN without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. These documents are available for ordering (hard copy) via the Naval Logistics Library https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

U.S. GENERAL SERVICES ADMINISTRATION

GSA Global Supply Catalog

(Copies of this document are available online at https://www.gsaglobalsupply.gsa.gov.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B16.5	-	Pipe Flanges and Flanged Fittings: NPS ½ through NPS 24, Metric/Inch Standard
ASME BPVC-I	-	Boiler and Pressure Vessel Code Section I-Rules for Construction of Power Boilers

(Copies of these documents are available online at www.asme.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQ Z1.4	-	Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at www.asq.org.)

ASTM INTERNATIONAL

ASTM A269/A269M	-	Standard Specification for Seamless and Welded Austenitic Stainless Steel Tubing for General Service
ASTM A351/A351M	-	Standard Specification for Castings, Austenitic, for Pressure-Containing Parts
ASTM A479/A479M	-	Standard Specification for Stainless Steel Bars and Shapes for Use in Boilers and Other Pressure Vessels

(Copies of these documents are available online at www.astm.org.)

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SAE INTERNATIONAL

SAE AMS2460 - Plating, Chromium

SAE J1086 - Numbering Metals and Alloys

(Copies of these documents are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.2.

3.2 Material.

3.2.1 General. Materials incorporated into these valve assemblies shall be as specified herein. Materials used in the design and construction of components shall be as shown in table I. Equivalent or better materials may be considered by the Technical Authority (see 6.6).

TABLE I. Materials.

Part	Material	Applicable documents
Body	Corrosion-resisting steel	ASTM A351/A351M (CF8 or CF8C); ASME BPVC-I
Bonnets	Corrosion-resisting steel	ASTM A351/A351M (CF8); or ASTM A479/A479M (UNS S30400); ASME BPVC-I
Nozzles and disks	Corrosion-resisting steel, hard surfaced	ASTM A479/A479M, MIL-R-17131(MIL-R-CoCr-A), and hard surfacing requirements of S9074-AQ-GIB-010/248 with S9074-AR-GIB-010/278
Threaded fasteners	Chromium-molybdenum vanadium steel	Bolts to grade B16 or better and nuts to grade 4 or better of MIL-DTL-1222
Pilot valve	Corrosion-resisting steel	ASTM A351/A351M (CF8); or ASTM A479/A479M (UNS S30400); ASME BPVC-I
Tubing and tubing couplings	Hard stainless steel	ASTM A269/A269M
NOTE: Unified Numbering System (UNS) alloy numbers in accordance with SAE J1086.		

3.2.1.1 Moving parts. Moving and threaded parts (excluding fasteners) made of corrosion-resisting steel that are in contact with similar material parts shall be plated to prevent galling or seizure (see SAE AMS2460).

3.2.1.2 Springs. Springs shall be made of non-corrosive materials that shall retain their properties under the service conditions specified herein (see 3.4.2.4 and 3.4.2.5).

3.2.1.3 Unloader valve. The unloader valve parts shall be made of the material specified in table I. Equivalent or better materials may be considered by the Technical Authority.

3.2.1.4 Pilot assembly. The pilot assembly may be subjected to continuous blowdown and, therefore, shall be constructed and its materials selected for withstanding full boiler water pressure and temperature in accordance with MIL-STD-777 and table I.

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3.2.2 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.2.3 Material identification and control. System fluid pressure retaining parts and the fasteners that provide the clamping force to retain these parts shall be subjected to strict material identification and control. Continuous traceability to the original heat or batch shall be maintained to ensure the identification of chemical and physical properties of these parts. These parts shall be marked to relate the part to the material chemical composition and physical properties documentation. Non-metallic materials and such items as setting or adjusting (threaded) parts with a ½-inch diameter or less shall be excluded from this requirement.

3.3 Construction. Construction of the safety valves shall be in accordance with 3.3.1 through 3.3.5.4.

3.3.1 Characteristics. Characteristics shall be in accordance with 3.3.1.1 through 3.3.1.3.

3.3.1.1 Configuration. The boiler safety valve system covered by this specification shall include the following general features, as defined hereinafter:

a. A safety valve (the unloader valve) for installation at the superheater outlet and the steam drum.

b. A hydraulic, non-steam flow pilot assembly for continuously sensing boiler drum pressure; for actuating, automatically, the lifting of the unloader valve for blowdown; and for repressurization for safety valve closure.

c. A boiler water storage canister, with filter, in the water supply line to the pilot valve assembly.

d. Means for manually actuating lifting (popping) and reseating of the safety valve.

e. The above items, other than the unloader valve, can be expected to be located away from the vicinity of the boiler to permit ease of maintenance.

Other appurtenances, to include a manually operated three-way valve and the self-contained piping between pilot assembly components, as necessary for satisfactory performance, shall be provided. Terminal piping and tubing connections for the sensing line, the water inlet line to the canister, and the actuating line shall be socket-welded ends located away from their units to allow attachment of piping, as necessary. These lines shall contain mechanical (takedown) fittings near the socket-welded ends.

3.3.1.2 Dimensions. NAVSHIPS 5000-S5103-841622 shows the general configuration of the unloader valve. The size of the safety (unloader) valve shall be the size of the inlet flange shown in table II.

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TABLE II. Dimensional and design data.

Style	Inlet flange type (ASME B16.5)	Inlet flange size (inch)	Outlet flange type (ASME B16.5)	Outlet flange size (inch)	Drain flange type (ASME B16.5)	Drain flange size (inch)	Flow rate (lb/hr)	Pop pressure (lb/inch²)	Steam temp. (°F)	Center to inlet (inch, ±1/16)	Center to outlet (inch, ±1/16)	Center to drain (inch, ±1/8)	Inlet to drain (inch, ±1/8)	Ship class
9	Class 1,500	2½	Class 300	4	Class 300	½	56,070	1,360	950	7¾	7	6 11/16	6 3/32	FF-1052
23	Class 600	2½	Class 300	3½	Class 300	½	39,500	705	870	6	6	6½	4 19/32	AS-39, LCC-19
25	Class 900	3	Class 300	6	Class 300	½	101,283	750	915	7	8	6¾	5 3/32	LHD-1
43	Class 900	4	Class 300	6	Class 300	½	169,200 or 171,400	765 or 775	520	7¾	8¾	7⅛	5⅛	LHD-1
51	Class 1,500	2½	Class 300	4	Class 300	½	54,900 or 55,600	1,385 or 1,405	590	7¾	7	6½	5 3/32	FF-1052
57	Class 600	2½	Class 300	3½	Class 300	½	43,300 or 44,000	720 or 730	520	6	6	6½	4 19/32	AS-39, LCC-19

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3.3.1.3 Pressure. Unless otherwise specified (see 6.2), safety valves shall be designed and constructed to one of the standard pressures specified in 3.3.1.3.1 through 3.3.1.3.3, as applicable.

3.3.1.3.1 600 pounds (nominal) standard. Valves for use on boilers having drum pressures of 362 to 675 lb/in² (for which the safety valve popping pressure may exceed 675 lb/in², but will not exceed 735 lb/in²) shall be constructed for a pressure of at least 735 lb/in² and an operating temperature of at least 875 °F.

3.3.1.3.2 900 pounds (nominal) standard. Valves for use on boilers having drum pressures of 676 to 975 lb/in² (for which the safety valve popping pressure may exceed 975 lb/in², but will not exceed 1,000 lb/in²) shall be constructed for a pressure of at least 1,000 lb/in² and an operating temperature of at least 950 °F.

3.3.1.3.3 1,500 pounds (nominal) standard. Valves for use on boilers having drum pressures of 676 to 1,600 lb/in² (for which the safety valve popping pressure may exceed 1,600 lb/in², but will not exceed 1,700 lb/in²) shall be constructed for a pressure of at least 1,700 lb/in² and an operating temperature of at least 1,000 °F.

3.3.2 Shipboard environmental conditions. Safety valves shall be designed and constructed to perform their required functions in a reliable manner when installed and operated onboard Navy ships under the following shipboard arrangements and environmental conditions, which shall be understood to prevail simultaneously and to occur in any combination.

3.3.2.1 Installation orientation. The valve shall be configured for installation in either the athwartships or the fore-and-aft orientation.

3.3.2.2 Atmosphere. The valve shall be constantly exposed to a marine environment that shall be understood to include extremes of ambient temperatures, high humidity in an at-sea atmosphere, and conditions conducive to electrolytic corrosion of dissimilar metals.

3.3.2.3 Shock. The total safety valve system shall be designed and constructed to withstand high-impact shock in accordance with MIL-DTL-901 for type A, grade A, class I equipment, while remaining functionally reliable and safe.

3.3.2.4 Vibration. The valve shall meet type I vibration requirements in accordance with MIL-STD-167-1.

3.3.2.5 Inclination. The valve shall be exposed to cyclic and possibly permanent ship inclinations of a ship in a seaway and shall perform reliably under the following inclination and motion requirements.

Permanent trim by bow or stern	5 degrees
Permanent list	15 degrees
Symmetrical rolling	45 degrees
Symmetrical pitching	10 degrees

These are displacements from the horizontal position. Trim and list as well as roll and pitch conditions shall be considered as occurring simultaneously.

3.3.3 Safety criteria. The design and construction shall be such that all procedures for operating, inspecting, maintaining, and adjusting the valve can be effected in a safe manner. To ensure maximum protection for operating and maintenance personnel against hazards and accidents, the valve shall, as a minimum, incorporate the following safeguards and protective features:

a. Exposed moving parts that could be a hazard to personnel shall be suitably covered by removable, expanded metal guards or similar sturdy material. Such guards shall not interfere with valve operation or maintenance.

b. Exposed edges shall be generously rounded and smooth to prevent cutting edges and sharp corners.

c. Safety valve components of heavy or unwieldy nature shall be mounted so that they shall not fall, topple, slide, or drop when their mounting fasteners are removed. Lifting the components for mounting and dismounting or assembly and disassembly shall constitute a separate action apart from removing the mounting fasteners.

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d. Means shall be provided to relieve all pressure-containing pockets without creating a hazard to personnel.

e. Venting or other means used when placing the safety valve system into use shall not constitute a hazard to personnel.

3.3.4 Maintainability. Ease of maintenance shall be a consideration. Proper regard shall be given to the fact that maintenance often may be performed under adverse conditions, with the ship at sea far from normal sources of supply and may be performed by maintenance personnel who may not be seasoned mechanics. Maintainability criteria shall include parts standardization, parts interchangeability, configuration control, accessibility for maintenance and repair, lifting, and portability.

3.3.4.1 Parts standardization. The design and construction shall standardize those components, fasteners, and items used in order to limit the number of sizes and kinds of items that are generally similar. For example, a design incorporating the same size bolts in two locations shall standardize on the same bolts, even if it shall mean upgrading the bolts at one of the two locations. Wherever practicable, standard and commercially available parts shall be used. Seals and gaskets shall be standardized to the maximum extent possible and shall be of a configuration and type commonly available from industrial supply houses.

3.3.4.2 Interchangeability. Identical components and parts, to include repair parts, furnished on the same contract or order or built to the same drawings shall be interchangeable as manufactured. Components that are functionally interchangeable shall, as far as practicable, be physically and mechanically interchangeable, as applicable. The unloader valve and pilot valve assembly shall be constructed for modular replacement of parts by ship's personnel. However, when two similar items are not interchangeable, functionally and strength wise, they shall not be physically interchangeable and shall differ significantly so that their non-interchangeability shall be apparent and shall ensure correct installation.

3.3.4.3 Configuration control. Components and their parts shall be constructed so that replacement parts subsequently constructed in accordance with the contractor's drawings can be replaced without additional machining, handfitting, or otherwise customizing (see 6.4).

3.3.4.4 Accessibility for maintenance and repair. These valves shall be constructed so that the degree of disassembly for maintenance and repair shall be held to a minimum. Wherever possible, disassembly and reassembly methods shall require only the use of standard tools. Cutting, welding, brazing, and allied processes shall not be required for access, disassembly, and reassembly associated with maintenance. If layup or maintenance on the valve or its boiler can adversely affect the safety valve system performance by loss of primary fluid, or other causes, the design and construction shall incorporate means for properly placing the system into service. Items such as petcocks shall not be used for this purpose.

3.3.4.5 Lifting and portability. Proper construction consideration shall be given to the maximum weight, bulk, size, and shape that maintenance personnel can lift and transport safely during repair onboard ship in accordance with MIL-STD-1472. Components that are not easily transported because of weight, shape, or finish shall be provided with padeyes, lifting lugs, or the equivalent.

3.3.5 Assembly processes.

3.3.5.1 Cleanliness. Components, castings, weldments, and machined parts, particularly their internal passages, shall be thoroughly cleaned and inspected for cleanliness prior to assembly. After assembly, internal passages shall be purged clean of all construction debris. During assembly and upon completion, openings where contaminants could enter shall be sealed or otherwise protected appropriately.

3.3.5.2 Alignment. Components and assemblies that are alignment-critical shall be positively held in place by dowel pins, fitted bolts, keys, rabbeted or tongue and groove joints, or other indexing means, as appropriate, to ensure proper alignment and prevent misalignment.

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3.3.5.3 Threaded parts. Threaded parts shall conform to FED-STD-H28 and FED-STD-H28/1, FED-STD-H28/2, FED-STD-H28/4, FED-STD-H28/6, FED-STD-H28/19, and FED-STD-H28/21. Screw threads used in the design and construction of these valve assemblies shall be in accordance with the Unified National Coarse (UNC) series class 2 or 3 fit, except for studs that shall be a class 3 fit in accordance with FED-STD-H28. Through bolting shall be the preferred method of joining threaded parts whenever possible. If not possible, then the use of studs shall be preferred to the use of capscrews or tapbolts. Bolt-studs and studs shall be sufficiently long to ensure that, when nuts are tightened, there shall be a minimum thread protrusion where the leading edge of the first thread on the bolt or stud end is flush with the face of the nut. Tapered pipe threads shall not be used as connections for piping or to components exposed to full boiler pressure.

3.3.5.4 Welding. Welding and allied processes, to include requirements for radiographic and magnetic particle inspection, shall be in accordance with S9074-AR-GIB-010/278.

3.4 Component construction.

3.4.1 Unloader valve.

3.4.1.1 Dimensions. Unloader valves shall meet the dimensional and design requirements for safety valves specified in NAVSHIPS 5000-S5103-841622 and in table II for the applicable style and class. Valve flanges shall be raised face. Valves shall be designed and constructed for modular replacement.

3.4.1.2 Valve nozzles (seat bushings) and discs. Valve nozzles shall be of the “full nozzle” or “through bushing” type for all operating pressures. If the valve disc is guided from beneath, the guide wings shall extend below the seat to a distance not less than one half of the valve diameter. Guides shall be supported in such a way as to ensure consistent operation and tightness and to ensure that steam force, while the valve is blowing, shall not cause binding or seizing of the valve. The unloader valve shall be a pressure seating design and construction in which the drum boiler water pressure acting on a piston keeps the valve tightly seated until valve lifting is initiated automatically by the pilot valve or by the manual means provided. During hydrostatic testing, water temperature shall not exceed 100 °F. The valve shall be constructed to attain full lift at a pressure not higher than 3 percent above the set pressure.

3.4.1.3 Valve gag. The valve shall contain a built-in gag that prevents unloader valve operation. A means shall be provided to permit a visible determination as to whether the valve is in the gagged or non-gagged position. A locking mechanism shall be provided to prevent gagging the valve without first opening the lock.

3.4.1.4 Radiographic and magnetic particle requirements. Valve body parts that may be subjected to full boiler steam pressure shall meet the radiographic inspection requirements of S9074-AR-GIB-010/278.

3.4.1.5 Drain connections. Three ½-inch nominal pipe size (NPS) drain connections shall be located at 90, 180, and 270 degrees from the outlet to completely drain the valve body of condensate. A ½-inch NPS flanged end drain nipple and two pipe plugs shall be provided loose with the valve in accordance with NAVSHIPS 5000-S5103-841622 and table II, herein.

3.4.2 Remote pilot assembly. The remote pilot assembly shall consist of those components remotely located from the unloader valve and shall include the remote pilot valve, the canister, the filter, the manually operated three-way valve, associated piping, and necessary appurtenances. Valve settings shall be controlled from the pilot assembly and changes in settings shall be made without the need to touch the unloader valve. The pilot assembly shall be made as compact as possible and shall be a separate, self-contained unit. The pilot assembly shall be located in and fixed to a rigid mounting plate in a metal cabinet that shall contain means for attachment to a bulkhead. The cabinet interior shall have ample space to permit access for adjustments and maintenance. A clear, well-prepared drawing of how the safety valve and remote pilot assembly shall be connected to the boiler shall be photo etched to a plate at least 12 by 8 inches and installed inside of the door of the pilot assembly cabinet. The plate shall also contain the valve information required in 3.6.d through h. The cabinet shall be painted with two coats of navy gray paint and shall contain means to permit locking.

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3.4.2.1 Remote pilot. The remote pilot shall be a multi-function valve and control system that shall:

a. Function using the boiler fluid as the hydraulic fluid.

b. Continuously sense the pressure in the boiler steam drum and act to maintain pressure on the unloader valve piston to keep it seated tightly during normal conditions.

c. Automatically cause the unloader valve to open fully when the set pop pressure is reached or exceeded.

d. Automatically cause the unloader valve to reseat at the set reseating pressure after blowdown.

e. Cause the unloader valve to fully open, blowdown, and reseat when actuated from a hand-operated lifting gear.

f. Incorporate other means to perform as specified herein.

3.4.2.1.1 Hand-easing gear. The pilot shall be provided with a linkage for the attachment of a hand-easing cable so that the three-way valve may be manually operated to actuate fully opening the unloader valve. The linkage shall be designed to fully open the three-way valve and return it to its normal operating position upon release of the force on the easing gear. To accommodate the shipboard hand-easing gear, fully opening the three-way valve shall require less than 3 inches of linkage travel, and between 150 and 180 pounds of lifting force on the linkage. In order to prevent damage, the linkage shall incorporate a device to prevent the three-way valve from exceeding its maximum travel.

3.4.2.1.2 Alternate gag. Provisions shall be made in the pilot valve to allow installation of a temporary clamping device that, when installed, shall prevent the safety valve from lifting, thus, acting as an alternative means of gagging the valve. Visible means shall be provided for showing clearly whether the valve is in the gagged or non-gagged position. Each pilot valve shall be supplied with a gag.

3.4.2.1.3 Locking mechanism. Locking mechanisms shall be provided that shall prevent changes in valve settings and prevent gagging the valve shut without first opening the locking mechanisms on the cabinet panel.

3.4.2.2 Piping and associated equipment. Piping for control systems shall be hard stainless steel tubing, UNS S30400 or S31600, in accordance with ASTM A269/A269M. Mechanical tubing connections shall be machine thread union type with O-ring seal. Piping, flanges, and fittings that may convey steam shall be in accordance with MIL-STD-777. Spiral wound gaskets may be of the inner and outer ring construction. Tapered pipe thread connections shall not be permitted.

3.4.2.3 Manual control valve. The manually operated three-way valve shall be capable of operation with boiler water up to 3,000 lb/in² and a temperature up to 590 °F, unless permitted otherwise by the Technical Authority. New three-way control valves shall be physically interchangeable with previously used valves.

3.4.2.4 Springs. The permanent set of the springs, 10 minutes after being released from a cold compression test which closes the spring solid, shall not exceed 0.015 inch. The springs shall not be compressed solid when the valve is at its maximum lift.

3.4.2.5 Spring range. Springs shall be constructed so that the popping and blowdown pressure can be varied over a range of approximately 10 percent of the ranges specified in 3.5.2.1 and 3.5.2.2, respectively.

3.4.2.6 Canister and filter. The canister and filter assembly shall remove particulate matter that could interrupt pilot valve operation. Flow capacity through the canister and filter shall be sufficient (even with a 20 percent reduction in filtration area caused by clogging) to permit blowdown and reseating as specified herein.

3.5 Performance.

3.5.1 Relieving capacity. The relieving capacity of the pilot operated safety valve shall be in accordance with the ASME BPVC-I or as specified (see 6.2), and shall be based upon 103 percent of popping pressure. The tolerance allowed for the actual capacity of each valve shall be -0 and +5,000 pounds per hour.

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3.5.2 Safety valve settings.

3.5.2.1 Popping pressure. The pilot actuator valve shall be designed and constructed to pop and open the superheater safety valve at a pressure of 8 percent greater than the highest working steam drum working pressure (120 percent overload). The low drum safety valve shall be set to pop at 2 percent of the highest working steam drum pressure higher than the pilot valve actuator valve setting. The high drum safety valve shall be set to pop at 1½ percent higher than the low drum valve. The valve shall be designed and constructed so that the popping pressure is adjustable and able to be set at any pressure within a range from 5 below to 5 percent over the popping pressure. Popping shall be achievable within ±5 lb/in² of the set popping pressure. Valve opening shall be immediate, positive, and characterized by a full opening pop action with no simmering. Prior to delivery, valves shall be set at the popping pressure specified (see 6.2) and verified by popping.

3.5.2.2 Blowdown. After opening, the valve shall remain fully open during blowdown until the pressure has dropped to the specified, preset reseating pressure; at that point, the valve shall close sharply and tightly without chattering or simmering. Unless otherwise specified (see 6.2), the reseating pressure shall be within the range of 3 to 4 percent below the popping pressure. Prior to delivery, valves shall be set for the reseating pressure specified (see 6.2) and verified by blowdown. Unless other allowable tolerances are specified (see 6.2), reseating shall be achievable within ±5 lb/in² of the set reseating pressure. Blowdown shall be within the allowable limits determined by the specified setting range of the valve. Valves shall be designed and constructed for the specified range of blowdown.

3.6 Marking. Safety valves shall be plainly marked. The markings shall be stamped or cast on the body or stamped on an identification plate permanently secured thereto that shall be in accordance with MIL-DTL-15024 and MIL-P-15024/5. Identification plates shall contain the following information:

a. Name or identifying trademark of contractor and designating number or symbol.

b. The contractor's model or type number and serial number, if applicable, that can positively identify the item.

c. The nominal size and nominal operating pressure for which the valve body is designed (preferably cast on the valve body; for example, 2½ – 600).

d. The authorized popping pressure.

e. The authorized reseating pressure.

f. The rated relieving capacity, as specified in 3.5.1.

g. Vertical lift of valve necessary to give rated relieving capacity.

h. MIL-DTL-24740, class and style (see 1.2.1 and1.2.2).

3.7 Special tools. The need for using special tools for adjustments or maintenance shall be minimized. Common tools and wrenches listed in the GSA Global Supply Catalog shall not be furnished when special tools are specified (see 6.2).

4. VERIFICATON

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on three safety valves of each new design and construction lot and style when a first article sample is required (see 3.1). This inspection shall consist of the examination and tests specified in table III (see 6.3).

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TABLE III. First article and conformance inspection.

Characteristics	Requirements	First article	Conformance
Material	3.2 through 3.2.3	4.4	---
Construction, safety, assembly, and marking	3.3 through 3.3.2.2, 3.3.3, 3.4 through 3.4.1.3, 3.4.2 through 3.4.2.6, 3.3.5 (all), 3.6, 3.7	4.4	---
Hydrostatic	3.4.1.2	4.5.1	4.5.1
Steam tightness	3.5.2	4.5.2	4.5.3
Shock	3.3.2.3	4.5.4	---
Vibration	3.3.2.4	4.5.5	---
Inclination	3.3.2.5	4.5.8.2	---
Popping and reseating pressures	3.5.2.1	---	4.5.6
Radiographic and magnetic particle inspection	3.4.1.4	4.5.7	4.5.7
Drain connections	3.4.1.5	4.4	---
Performance	3.5	4.5.8 through 4.5.8.2.1	---
Maintainability	3.3.3, 3.3.4 (all), 3.3.5 (all), 3.7	4.5.9	---

4.3 Conformance inspection. Conformance inspection shall consist of the tests specified in table III on a sample (see 4.3.1).

4.3.1 Sampling for conformance. Boiler safety valves (including components) of the same size, class, and style presented for delivery at one time shall be considered a lot for purposes of conformance inspection. A random sample of safety valves shall be selected from each lot for the conformance inspections specified in table III in accordance with ANSI/ASQ Z1.4 at inspection level II and the Acceptance Quality Limit (AQL) specified (see 6.2).

4.3.2 Action in event of failure. If a valve fails a requirement, the valve shall be segregated. Corrective action shall be as specified (see 6.2).

4.3.2.1 Re-examination. If a sample fails the examination, each valve in the lot shall be examined to identify the deficiencies, the deficiencies shall be satisfactorily corrected, and the valve shall pass the examination of 4.4 to determine acceptability. Defects shall be as specified in table IV.

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TABLE IV. Classification of defects.

Categories	Defects
Critical	None defined
Major	
101	Valve not style, class, or size specified
102	Valve not complete; component parts missing
103	Component parts not new; evidence of reclaimed, used, or salvaged components
104	Evidence of use of unauthorized materials or repairs
105	Threaded parts nonconforming; not specified class, fit; evidence of burring, crossing, stripping
106	Threaded projection from outer face less than required
107	Springs not of noncorrosive material
108	No stop provided to fix position of valve after adjustment
109	Valve nozzles not welded into valve body; not “full nozzle” or “through bushing” type, as specified
110	Lock mechanism does not prevent altering spring adjustment without opening lock
111	Hand-lifting gear not provided
112	Drain connection size and location nonconforming
113	Similar parts not interchangeable
Minor	
201	Markings nonconforming; not stamped or cast on body, or stamped on identification plate permanently secured to body; not permanent; not legible; data incomplete
202	Specified repair parts not furnished
203	Specified special tools and accessories not furnished
204	Specified drawings not furnished; not specified type; incomplete, incorrect
205	Specified technical manuals not furnished; incomplete, incorrect

4.3.2.2 Re-testing. If a sample had previously failed to pass a test, each valve in the lot shall have the deficiencies corrected and shall pass the tests in 4.5.

4.4 Visual examination. After completion of first article tests, the valves shall be examined for signs of weakness, incipient cracks, or other disruptions which might harm personnel or create other hazards. The visual examination shall include verification of dimensions and materials, checking for cleanliness of parts and missing parts, and verifying correct assembly and alignment.

4.5 Test procedures. Shock testing (see 4.5.4) and vibration testing (see 4.5.5) shall be conducted prior to the other tests and examinations required.

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4.5.1 Hydrostatic test. Apply water pressure of 1½ times the popping pressure of the highest set pressure of the valves to both the steam inlet and hydraulic line connections of the valve for a period of three minutes (see 3.4.1.3). Water droplets may appear at the seating surface but should not run down the side of the nozzle.

4.5.2 Steam tightness (first article). The steam boiler used to supply steam for testing shall be of sufficient capacity to cover the full pressure drop necessary to achieve blowdown requirements for the valve concerned. The procedure outlined below shall be considered the standard for determining steam tightness. Valves tested shall be free from preservative prior to testing. After a warming-up period on the test stand, the valve shall be popped, reseated, and adjusted to the proper popping pressure and blowdown within the tolerances specified in 3.5.2.1 and 3.5.2.2. The valve shall then be popped by steam 50 times, after which a steam-tight cover shall be affixed to the exhaust flange and the line connected to a cooling condenser. The condensate from the condenser shall be collected for measurement. Measurement shall be conducted as follows:

a. Steam pressure shall be raised to 90 percent of set pressure.

b. The condensate shall be measured from the above system for a period of 30 minutes to determine if steam tightness requirements have been met. Any leakage shall be measured from the pilot panel. There shall be no visible steam leakage between the valve spindle and its guide. The measured leakage rate (total of the condensed leakage from the valve and any boiler water leakage from the pilot assembly) shall be not greater than 1.12 ounces/hour (33 cubic centimeters/hour) from either the safety valve or the pilot valve.

4.5.3 Steam tightness (conformance). The steam source used to supply steam for testing shall be of sufficient capacity to cover the full pressure drop necessary for the valve concerned. The procedure outlined below shall be considered the standard for determining steam tightness for delivery. Valves tested shall be free from preservative prior to testing. After a warming-up period on the test stand, the valve shall be set at the required settings within the tolerance specified in 3.5.2.1 and popped by steam five times, after which a steam-tight cover shall be affixed to the exhaust flange and the line connected to a cooling condenser. The condensate from the condenser shall be collected for measurement. Measurement shall be conducted as follows:

a. Steam pressure shall be raised to 90 percent of set pressure.

b. The condensate shall be measured from the above system for a period of 15 minutes to determine if steam tightness requirements have been met. Any leakage from the pilot valve assembly panel drains shall be measured.

The leakage rates above shall be the total of the condensed leakage from the valve and boiler water leakage from the pilot valve assembly. Steam tightness acceptance shall be based on the same criteria as in 4.5.2.b.

4.5.4 Shock. High-impact shock tests for the unloader valve and the pilot assembly shall be conducted in accordance with type A of MIL-DTL-901 for grade A, class I equipment. The valve shall be considered to have met the high-impact shock requirements if the following conditions have been met:

a. No breakage of any part, including accessories and bolting, shall have occurred. Examination shall reveal no incipient cracks or signs of weakness that could result in failure if operated.

b. The valve shall have performed and shall continue to perform without maintenance or adjustment.

c. There shall have been no dislocation of components or appurtenances which could render the valve unsafe for continued service.

4.5.5 Vibration. Type I vibration tests for the pilot valve assembly and superheater (unloader) valve shall be conducted in accordance with MIL-STD-167-1. The valve shall be considered to have met the vibration test requirements if it continues to perform accurately during and after the vibration tests and shows no evidence of weakness or disarrangement.

4.5.6 Popping and reseating pressures. Each valve shall be popped at three popping pressures to cover the setting range of the spring (see 3.5.2.1). The settings shall be chosen to ensure that the range of setting shall be obtained. No simmering shall be permitted at any set pressure. Valve closure shall be clean and sharp. Failure shall be defined as follows:

a. Full popping range not covered.

b. Lifting and reseating pressure nonconforming.

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c. Time lag between lifting and reseating on pilot and low drum valves nonconforming.

d. Popping and reseating are not clean and sharp.

After completion of these tests and prior to packaging, each valve shall be adjusted to the settings required by the contract or order (see 3.5.2 and 6.2) and verified by test.

4.5.7 Radiographic and magnetic particle inspection. Portions of the valve that may be subjected to full boiler pressure while in service shall be subjected to radiographic and magnetic particle inspection in accordance with S9074-AR-GIB-010/278. There shall be no evidence of pockets, porosity, occlusions, slag, hard spots, or discontinuities.

4.5.8 Performance. The valves shall be set to pop and reseat at the popping and reseating pressures specified (see 6.2) when the boiler is under static load (no steam flow through the main steam outlet). Superheater safety valves shall be set with steam having superheat not greater than 50 °F. After such setting, the boiler shall be brought up to the rating necessary to hold the valve open and to obtain the specified steam temperature at the superheater outlet. This rating shall be held for not more than 1 hour before popping each valve. To evaluate temperature effect on performance, each valve shall then be popped with the boiler under running load (steam flowing through the main steam outlet), held open for 2 minutes, and then reseated. The popping and reseating pressures under this condition shall not vary from the settings specified (see 6.2) by more than ±1 percent of the reseating pressure. Immediately after repeating, the valve shall be popped again and allowed to reseat. The time interval before popping again shall be only that required to control the boiler pressure. The variations in popping pressure under this condition from the specified popping pressure shall not exceed 1½ percent of the specified reseating pressure, and the variation of the reseating pressure from the specified reseating pressure shall not exceed 1 percent of that pressure. Reduction in blowdown within the above tolerances shall not affect the proper functioning of the valve.

4.5.8.1 Sequence time. In addition to the tests specified in 4.5.8 (or in conjunction therewith), runs shall be made to ascertain the time lag between popping and reseating of the pilot section and the lifting and reseating of the unloader valve. Under static load, with the main stop secured, the interval between opening of the pilot valve and lifting of the unloader valve shall not exceed 2 seconds. Similarly, the interval between reseating of the pilot valve and reseating of the unloader valve shall not exceed 2 seconds. Under running load with specified steam temperature at the superheater outlet, the time lag shall not exceed 2 seconds in either case.

4.5.8.2 At-sea performance test. After the valve has passed all other first article tests, arrangements shall be made through the Technical Authority to install a minimum of two valves on a test ship. The valves shall be set to required pressure and popped by steam to confirm the setting. The valves shall be left on the ship for a period of at least 9 months to experience normal operation. The valves will then be inspected and evaluated for performance history, maintenance requirements, and reliability by the Naval Surface Warfare Center Philadelphia Division (NSWCPD). Corrective action shall be as specified (see 6.2).

4.5.8.2.1 At-sea performance test acceptance criteria. The purpose of the at-sea performance test is to confirm that the new system maintains its availability to function properly over a prolonged period of time in the shipboard environment. Valve performance shall be considered acceptable if:

a. Valve settings are maintained within ±5 lb/in² of original required settings.

b. Valve functions properly; lifts and reseats without chatter or simmer.

c. When retested in the laboratory after shipboard service, valve displays no visible leakage on ship and meets the requirements of 4.5.2.

d. No malfunctions are experienced during the testing period or all malfunctions have been corrected and the system operated satisfactorily as indicated above for a 6-month period or longer as prescribed by the Technical Authority.

e. Replacement of parts caused by wear or imminent malfunction shall not be required after the shipboard test. Allowances will be made for normal wear.

4.5.9 Maintainability. The maintainability of the safety valve system shall be demonstrated by the successful completion of all examinations and tests. The maintenance procedures, safety of execution, parts standardization and interchangeability, accessibility, lifting requirements, and assembly processes shall be verified during first article testing. The demonstration procedures in MIL-HDBK-470 may be used.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The boiler safety valves covered by this specification are intended for installation at the superheater outlet or on the steam drum with the pilot assembly located remotely from the boiler and unloader valve. This provides features for ease of maintenance, shipboard level repair capability for satisfactory replacement of moving parts, high service time to leakage rate ratio, simplified ease of setting features, cold pilot setting ability, and a built-in gagging feature. The valve uses boiler fluid as the working fluid and operates completely without any auxiliary means of power such as air or electricity. These valves have safety and construction characteristics in excess of similar commercial items, making them suitable for use aboard Naval vessels. Current stock of boiler safety valves may be used until depleted; future requisitions for boiler safety valves should refer to the current version of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Class and style of valve (see 1.2).

c. When first article inspection is required (see 3.1).

d. Design pressure (highest working steam drum), if other than specified (see 3.3.1.3). (Note that the pressure class of the inlet flange, as shown in table II, does not necessarily coincide with the valve design pressure.)

e. Material of bolt-studs and studs, if other than specified (see 3.3.5.3).

f. Relieving capacity, if other than specified (see 3.5.1).

g. The popping pressure setting (see 3.5.2.1).

h. The reseating pressure setting and tolerances, if other than specified (see 3.5.2.2).

i. Special tool requirements (see 3.7).

j. AQL of 1.0 (see 4.3.1).

k. Corrective action requirements in the event of valve failure (see 4.3.2).

l. Corrective action requirements in the event of at-sea performance test performance, maintenance, reliability, or material problems (see 4.5.8.2).

m. Packaging requirements (see 5.1).

n. Documentation requirements (see 6.4).

o. Repair parts requirements (see 6.5).

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6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the items should be a pre-production sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.2. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Recommended documentation. The following documentation should be requested as specified (see 6.2):

a. Test and material reports

b. Technical manuals

c. Drawings

d. Provisioning technical documentation

6.5 Repair parts. When specified (see 6.2), repair parts required for maintenance should be furnished.

6.6 Technical Authority. The Technical Authority has the authority, responsibility, and accountability to establish, monitor, and approve technical standards, tools, and processes in conformance to higher authority policy, requirements, architectures, and standards. NAVSEA 05S may be contacted to determine the appropriate Technical Authority for any given requirement related to equipment and systems of U.S. Navy ships and submarines.

6.7 Cross-reference of classifications and substitutability data. Class 2 safety valves for 1,500 lb/in² service have been redesignated as class 3 in revision A of this specification. As such, class 3 safety valves for 1,500 lb/in² service in accordance with MIL-DTL-24740A are substitutable for class 2 safety valves for 1,500 lb/in² service in accordance with MIL-V-24740.

6.8 Subject term (key word) listing.

At-sea performance

Blowdown

Inclination

Popping pressure

Relieving capacity

Shut down and start up

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

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